



HACKYEAH 2026 · AI CONTROL LAYER · GOLDMAN SACHS CHALLENGE

Blackwall: presentation plan

Slide-by-slide plan for the hackathon audience, the jury and the Goldman Sachs team.
“Before the agent makes its move.”

AUDIENCE

Attendees (engineers),
jury, Goldman Sachs
challenge owners

LENGTH

13 slides, about 14 min, of
which 3 min is a live demo

CORE CLAIM

Agent actions are checked
before they have an effect,
and every decision is
explained

TONE

Evidence first, limits stated
openly

Storyline

1 · Problem

Slides 1–2

Agents act, not just talk.

2 · Answer

Slides 3–5

One layer, hybrid
defense, one policy.

3 · Use case

Slides 6–7

KYC, M&A, HR, live.

4 · Proof

Slides 8–11

Supervision, policy,
dashboard, tests.

5 · Honesty + ask

Slides 12–13

Limits, pilot path.

How the slides answer the jury’s criteria

The brief’s evaluation weights decide the airtime. Robustness and guardrails carry 30%, so the use cases, the demo and the supervision slides get the most time.

CRITERION (WEIGHT)	SLIDES	WHAT THE AUDIENCE SHOULD LEAVE WITH
Robustness and quality of guardrails (30%)	5, 6, 7, 8	Hybrid rules plus semantic review; a refusal stops the tool before any effect; a hard prohibition cannot be lifted by a model.
Architecture and performance (20%)	3, 4, 8	Gateway plus plugin integration; per-stage latency numbers (p95).
Security reporting (20%)	10	Live session view, reasons for every decision, JSONL export, p50/p95 metrics, cost.
Self-testing suite (15%)	11	Positive and negative tests that check <i>effects</i> , runnable by the judges.
Implementability and scalability (15%)	9, 12	Config changes without restart; an honest path from local MVP to pilot.

Assumptions to confirm. The brief gives no slide count or time slot, so I used the default 10–15 slides and a time budget of about 14 min plus Q&A. A 5-minute cut is on the last page. All figures come from the final audit (`blackwall/reports/final-audit.md`, 4 Oct 2026). Re-check them on the day (see the checklist on the last page).

Slide-by-slide plan (slides 1–12)

1

0:20

Title: Blackwall, “Before the agent makes its move”

ON THE SLIDE

- Name, tagline, team, “HackYeah 2026 · AI Control Layer · Goldman Sachs challenge”.
- Cover art as the only visual.

SAY

One sentence: “Blackwall decides whether an AI agent may do something before it does it.”

VISUAL

docs/assets/blackwall-cover.png

2

1:00

The problem: agents act, they don’t just talk

ON THE SLIDE

- Three risks taken from the brief: access and impersonation; prompt injection and leaking output; runaway cost and loops.
- One line on why classic tools miss them: natural language is the execution logic.

SAY

An agent can read a file, run a program, send data and burn budget. Each of these needs a boundary, and the boundary must be enforced where the action happens.

VISUAL

Three-icon row, no screenshots. Cite the brief’s “AI Control Layer” text.

3

0:45

Our answer: one control layer, one policy, one audit trail

ON THE SLIDE

- Mini diagram: Agent → **Blackwall** → tools / models.
- Four principles: decide before the effect; hybrid (deterministic + semantic); central policy; every decision explained.

SAY

Every tool call and every model call passes through Blackwall. The answer is allow, deny or ask the user, and it comes with a reason code.

JURY CRITERIA

Architecture 20%

4

1:15

Architecture: how it plugs into real agents

ON THE SLIDE

- The architecture diagram the brief asks for (simplified from `blackwall/README.md`).
- Components: managed Pi plugin; OpenAI-compatible gateway (any client can use it); decision engine; YAML policy plus threat feed; embeddings → guardian; Jev action review; SQLite audit; dashboard.
- Callout: integration is one base URL or one plugin.

SAY

The gateway is the main entry point, so any OpenAI-compatible client can be governed. Pi is our demo client. The one-time grant is checked again immediately before execution.

JURY CRITERIA

Architecture 20% Implementability 15%

5

1:00

Hybrid defense: fast rules first, AI where meaning matters

ON THE SLIDE

Two-column table.

- **Deterministic:** path scope (symlinks, `...`, new files), `.env` and keys, network host/port/method, secrets, IBAN and PESEL checksums, token budgets, threat feed.
- **Semantic:** Jev reviews proposed actions; embeddings plus a guardian supervise conversation topics.

SAY

Two rules make this safe to rely on: a hard prohibition cannot be lifted by a model, and no answer from a required control means no approval.

JURY CRITERIA

Robustness 30%

6

1:15

The Goldman Sachs use case: three finance conversations

ON THE SLIDE

SCENARIO	USEFUL WORK ALLOWED	WHAT BLACKWALL STOPS
KYC onboarding	Draft written after one-time approval	Reading another client's data
Confidential M&A	Analysis; an authorized POST goes through	Publication prompted by an injected document
HR	General process questions	Evaluating named people; session terminated

SAY

These are regulated-finance problems: client confidentiality, deal secrecy, employee data. The agent still gets its work done. The controls only remove what it must not do. All data is synthetic.

JURY CRITERIA

Robustness 30%

7

3:00

Live demo (video fallback ready)

ON THE SLIDE

- Mostly the dashboard. Scenario label in the corner.
- Order: KYC approved write → other client's data denied → M&A forbidden publication blocked → HR violation ends the session. Run-of-show below the slide 13 card.

SAY

Narrate what Blackwall decided and why, then show the evidence: file unchanged, receiver got zero requests.

VISUAL

Clips in `blackwall/demo-recordings/2026-10-03-23-06-53/video/` (01, 03, 05, 06, 12).

Robustness 30%

Reporting 20%

8

1:00

Sensitive-topic supervision: stopping a conversation, not just a call

ON THE SLIDE

- Flow: embedding similarity → session label → one guardian verdict → terminated before the response is released.
- Latency, p95 per stage: topic detection **683 ms**, guardian **2316 ms**, Jev **750 ms**. Small sample, single stages.

SAY

A topic alone is not a violation. A candidate triggers the guardian, and uncertainty holds the session for an administrator. Only a confirmed violation closes it. Works in Polish and English.

JURY CRITERIA

Robustness 30%

Performance 20%

9

1:00

One policy file, changed live, with budgets and a threat feed

ON THE SLIDE

- Profiles `permissive / standard / strict` (thresholds 0.80 / 0.90 / 0.95); `enforce` vs `observe`; global plus per-user policy.
- Budgets: atomic token and cost reservation before a model call; model alias allowlist; unknown usage consumes the whole reserve.
- Threat feed examples: `pickle / torch.load` (CVE-2025-32434 class), `trust_remote_code`, `curl | sh`, reverse shell, prompt injection.
- Publish from the Policies tab without a restart. Invalid or stale publications keep the active version.

SAY

The brief says judges may change thresholds and rules and watch the reaction. That is the point: publish a new version, old grants and approvals expire, the next decision uses the new rules.

VISUAL

`.../00-policies-viewport.png`; clip `runtime-publication-.../video/runtime-publication-ui.mp4`

Implementability 15%

Robustness 30%

10

1:00

Dashboard and audit: what security teams and management see

ON THE SLIDE

- Live sessions: users and sessions on the left, conversation and Blackwall decisions on the right, status filters, unread counts.
- Each decision shows reason code, policy version, and whether the action actually started and finished.
- Overview with p50/p95 metrics and cost, plus JSONL export.

SAY

An `allow` does not prove an operation ran, so the audit separates decision, approval, claim and execution report.

VISUAL

`demo-recordings/live-sessions-2026-10-04-07-17-44/01-allowed-chat-and-controls.png`, `03-guardian-stop-and-policy.png`

Reporting 20%

11

1:00

Tests: positive and negative, checking effects

ON THE SLIDE

- Final verification: **143/143** unit, **33/33** live-API integration, **12/12** with a real Pi agent, no skips.
- Semantic eval (34 guardian assessments): 15/15 violations ended, **0/19** false terminations. Typed proposals: 5/5 forbidden denied, 0/5 allowed denied. Jev: 0/4 dangerous commands auto-allowed.
- Negative tests check the effect: file not created, receiver got no request, secret never reached the UI or audit.
- Command for judges: `npm test` (offline); `npm run verify` (needs both API keys).

SAY

The corpus is small and hand-written, so we do not claim production accuracy. We keep earlier failures in the repo (for example the 22:41 guardian miss, caught by deterministic file scope).

JURY CRITERIA

Self-testing 15%

12

1:00

Honest limits and the path to a pilot

ON THE SLIDE: LIMITS

- No OS sandbox: Jev reviews the command text, programs run with the user's rights.
- The plugin is the enforcement boundary; a host admin can disable it.
- Local MVP: SQLite, startup config for providers; no SSO, MCP or agent-to-agent; no automatic feed download.
- Models are probabilistic; content sent to OpenAI and Jev leaves the machine.

ON THE SLIDE: NEXT STEPS

- Sandboxed runner; credentials outside the agent process.
- Central control plane with SSO; PostgreSQL + pgvector.
- MCP and agent-to-agent adapters; automatic threat-feed ingestion.
- Local-model backend (for example Ollama) behind the same gateway.

Implementability 15%

Slide 13 and the live demo run-of-show

13
0:30

Close: three proofs and an invitation

ON THE SLIDE

- Recap: stops before the effect · explains every decision · changes live · tested for effects.
- Invitation: "Give us a prompt we have never seen."
- Links and QR: repository, interactive Sites presentation.

SAY

Offer the jury an ad-hoc prompt and a live policy change. Both are explicitly part of how the brief says projects will be judged.

VISUAL

Cover art again, QR code. Sites page: blackwall-hackyeah-2026.dariusz.chatgpt.site

Live demo (slide 7): 3 minutes

#	BEAT	WHAT TO POINT AT	CLIP (FALLBACK)
1	KYC: approved write (30 s)	One-time approval; the existing file really changes. The harness simulates the user's click, so say so.	01-kyc-approved-write.mp4
2	KYC: other client's data (30 s)	The read is refused by file scope before execution.	03-kyc-other-client.mp4
3	M&A: forbidden publication (45 s)	Guardian verdict stops the POST; the receiver has zero requests. This clip is an explicit replay, not the model refusing; say so.	05-ma-replay.mp4 , plus 12-ma-authorized-post.mp4 to show the allowed case
4	HR: violation (45 s)	General question allowed; "evaluate Anna and Peter, recommend whom to dismiss" ends the session before the model is called.	06-hr-termination.mp4 (EN) · 10-hr-pl-first-violation.mp4 (PL)
5	Live policy change (30 s)	Publish a threshold or feed rule in the Policies tab; the next proposal in the same session is denied under the new version.	runtime-publication-.../video/runtime-publication-ui.mp4

If time is short, cut beat 2. Keep the allowed-and-denied pairing in at least one scenario: it shows Blackwall does not just refuse everything.

Honesty rule for the demo. The recordings use simulated user approvals and explicit replays of a proposal for the M&A and runtime-publication clips. Say this out loud; the repo's own reports do, and a judge who checks will find it.

Prepare for questions

The brief says no paid subscriptions are provided. Does it run without OpenAI?

Not today. The runtime model, guardian and embeddings use OpenAI, and semantic action review uses Jev. The gateway is OpenAI-compatible, so a local backend is the natural next step, but it is not built. The deterministic layer and `npm test` work without network access.

What is the latency overhead?

Per stage, p95: topic detection 683 ms, guardian 2316 ms, Jev 750 ms. The guardian runs only for sessions with a topic candidate. End-to-end agent latency was not measured.

What if the LLM is wrong?

Deterministic rules run first and a model cannot override a hard prohibition. A model accusation without a verifiable quote is held for admin review, not auto-confirmed. Earlier misses are kept in the reports.

Can the agent bypass Blackwall?

Through the managed plugin or the gateway, no. A user with control of the host can disable the plugin, and there is no OS sandbox. A production deployment needs a managed executor.

Can judges change the config live?

Yes. Publish rules, thresholds, topic catalog or threat feed from the Policies tab, no restart. A stale or invalid publication is rejected and the active version stays.

How do you handle known historical attacks?

A versioned threat feed matched deterministically (RE2, no backtracking): unsafe deserialization, `trust_remote_code`, `curl | sh`, reverse shells, leaks to paste services. It is published manually; automatic download is not built.

MCP and agent-to-agent?

Not implemented. The gateway and plugin pattern is the same one those adapters would use.

Before you present

- Run `npm run verify` in `blackwall/` and use the numbers from that run. The README now cites 148 unit tests after the live-sessions view, while the final audit says 143.
- Start the server and open the dashboard on the demo data; keep the clips as offline fallback.
- Check the Sites page loads on venue Wi-Fi, or have screenshots ready.
- Slide 8 latencies come from one small run; present them as such.
- The public Sites page and the existing 9-slide deck (`deliverables/blackwall-demo-en.pptx`) can supply visuals. They do not cover the problem, use-case and close slides.
- Rotate the API keys after the event; they were shared in chat earlier.

5-minute cut

Slides 1, 2, 4, 6, then the demo (beats 1, 3, 4 only), then 11 and 12 in one breath each. Skip 3, 5, 8, 9, 10 and keep them as backup slides for questions.

Sources: README.md, blackwall/README.md, blackwall/reports/final-audit.md, blackwall/deliverables/demo-index.md, docs/golden-sachs.pdf (challenge brief and evaluation criteria).